

Chapter 11

Cluster Analysis

11.1 The Problem

Cluster analysis is a set of tools and methods for building groups (clusters) from multivariate data objects. The aim is to find groups with homogeneous properties out of heterogeneous large samples. The groups or clusters should be as homogeneous as possible and the differences between different groups as large as possible. Cluster analysis can be divided into two fundamental steps.

1. *Choice of a proximity measure:*

One checks for each pair of observations (objects) the similarity of their values. A similarity (proximity) measure is defined that measures the "closeness" of the objects. The "closer" they are the more homogeneous they are.

2. *Choice of group-building algorithm:*

On the basis of the proximity measures the objects are classified into groups so that differences between groups become large and observations in a group become as close as possible.

Cluster analysis is applied for example in marketing for the selection of test markets. Other applications include the classification of companies according to their organization structures, technologies and types. In psychology cluster analysis is used to find types of personalities through questionnaires. In archaeology it is applied to classify art objects in different time periods. Other scientific branches that use cluster analysis are medicine, sociology, linguistics and biology. In all these problems one analyzes a heterogeneous sample of objects with the aim to identify homogeneous subgroups of this sample.

- * Cluster analysis is a set of tools and methods for building groups (clusters) from multivariate data objects.
- * The methods are usually divided into two fundamental steps: The choice of a proximity measure and the choice of group-building algorithm.

Summary 11.1.

11.2 The Proximity between Objects

The starting point of cluster analysis is the data matrix $\mathcal{X}(n \times p)$ with n measurements (objects) of p variables. The proximity (similarity) between objects is measured by a distance matrix $\mathcal{D}(n \times n)$

$$\mathcal{D} = \begin{pmatrix} d_{11} & d_{12} & \dots & \dots & \dots & d_{1n} \\ \vdots & d_{22} & & & & \vdots \\ \vdots & \vdots & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ d_{n1} & d_{n2} & \dots & \dots & \dots & d_{nn} \end{pmatrix}. \quad (11.1)$$

The matrix $\mathcal{D}$ contains measures of similarity or dissimilarity between the n objects. If the values d_{ij} are distance values they measure the dissimilarity: The greater the distance the less similar the objects are. If the d_{ij} are proximity measures it is the other way around: the greater the proximity value the more similar the objects are. A distance matrix could for example be defined by the L_2 -norm: $d_{ij} = \|x_i - x_j\|_2$ where x_i and x_j denote as usual the rows of the data matrix $\mathcal{X}$. Distance and similarity are of course dual. If d_{ij} is a distance then $d'_{ij} = \max_{i,j} \{d_{ij}\} - d_{ij}$ is a proximity measure.

The nature of the observations plays an important role in the choice of proximity measure. Nominal values (like binary variables) lead in general to proximity values whereas metric values lead (in general) to distance matrices. We first present possibilities for $\mathcal{D}$ in the binary case and then consider the continuous variable case.

Similarity of objects with binary structure

In order to measure the similarity between objects we always compare pairs of observations (x_i, x_j) ; $x_i^T = (x_{i1}, \dots, x_{ip})$, $x_j^T = (x_{j1}, \dots, x_{jp})$, $x_{ik}, x_{jk} \in \{0, 1\}$. We have obviously four cases:

$$\begin{aligned} x_{ik} &= x_{jk} = 1, \\ x_{ik} &= 0, x_{jk} = 1, \\ x_{ik} &= 1, x_{jk} = 0, \\ x_{ik} &= x_{jk} = 0. \end{aligned}$$

Define now

$$a_1 = \sum_{k=1}^p \mathbf{I}(x_{ik} = x_{jk} = 1),$$

Name	δ	λ	Definition
Jaccard	0	1	$\frac{a_1}{a_1 + a_2 + a_3}$
Tanimoto	1	2	$\frac{a_1 + a_4}{a_1 + 2(a_2 + a_3) + a_4}$
Simple Matching (M)	1	1	$\frac{a_1 + a_4}{p}$
Russel and Rao (RR)	-	-	$\frac{a_1}{p}$
Dice	0	0.5	$\frac{2a_1}{2a_1 + (a_2 + a_3)}$
Kulczynski	-	-	$\frac{a_1}{a_2 + a_3}$

Table 11.1. The common similarity coefficients.

$$\begin{aligned}
 a_2 &= \sum_{k=1}^p \mathbf{I}(x_{ik} = 0, x_{jk} = 1), \\
 a_3 &= \sum_{k=1}^p \mathbf{I}(x_{ik} = 1, x_{jk} = 0), \\
 a_4 &= \sum_{k=1}^p \mathbf{I}(x_{ik} = x_{jk} = 0).
 \end{aligned}$$

Note that each a_ℓ , $\ell = 1, \dots, 4$ depends on the pair (x_i, x_j) .

The following proximity measures are used in practice

$$d_{ij} = \frac{a_1 + \delta a_4}{a_1 + \delta a_4 + \lambda(a_2 + a_3)} \quad (11.2)$$

where δ, λ are weighting factors. According to the weighting factors we have the following similarity measures.

These measures (Table 11.1) provide alternative ways of weighting mismatchings and positive (presence of a common character) or negative (absence of a common character) matchings. In principle we could also consider the Euclidian distance. Disadvantage of this distance though is that it measures the observation 0 and 1 in the same way. If $x_{ik} = 1$ denotes, say, knowledge of a certain language then the contrary $x_{ik} = 0$ i.e. not knowing this language should eventually be treated differently.

EXAMPLE 11.1 Let us consider a binary data set computed from the car data set (Table A.6). We define the new binary data by

$$y_{ik} = \begin{cases} 1 & \text{if } x_{ik} > \bar{x}_k, \\ 0 & \text{else,} \end{cases}$$

for $i = 1, \dots, n$; $k = 1, \dots, p$. This means that we transform the observation of the k -th variable to one if it is larger than the mean value of all observation of the k -th variable. Let us consider only the data points 17-19 (Renault 19, Rover and Toyota Corolla) which lead to (3×3) distance matrices. The *Jaccard* measure gives the similarity matrix

$$D = \begin{pmatrix} 1.000 & 0.000 & 0.333 \\ & 1.000 & 0.250 \\ & & 1.000 \end{pmatrix},$$

the *Tanimoto* measure yields

$$D = \begin{pmatrix} 1.000 & 0.231 & 0.600 \\ & 1.000 & 0.455 \\ & & 1.000 \end{pmatrix},$$

whereas the *Single matching* measure gives

$$D = \begin{pmatrix} 1.000 & 0.375 & 0.750 \\ & 1.000 & 0.625 \\ & & 1.000 \end{pmatrix},$$

Distance measures for continuous variables.

A wide variety of distance measures can be generated by the L_r -norms, $r \geq 1$,

$$d_{ij} = \|x_i - x_j\|_r = \left\{ \sum_{k=1}^p |x_{ik} - x_{jk}|^r \right\}^{1/r}. \quad (11.3)$$

Here x_{ik} denotes the value of the k -th variable on object i . It is clear that $d_{ii} = 0$, $i = 1, \dots, n$. The class of distances (11.3) for different r measures the dissimilarity with different weight. The L_1 -metric for example gives less weight to outliers as the L_2 -norm (Euclidean norm). It is common to consider the squared L_2 -norm.

EXAMPLE 11.2 Suppose we have $x_1 = (0; 0)$, $x_2 = (1, 0)$ and $x_3 = (5, 5)$ then the distance matrix for the L_1 -norm is

$$D_1 = \begin{pmatrix} 0 & 1 & 10 \\ 1 & 0 & 9 \\ 10 & 9 & 0 \end{pmatrix},$$

and for the squared L_2 - or Euclidean norm

$$D_2 = \begin{pmatrix} 0 & 1 & 50 \\ 1 & 0 & 41 \\ 50 & 41 & 0 \end{pmatrix}.$$

One sees that the third observation x_3 receives much more weight in the squared L_2 -norm than in the L_1 -norm.

An underlying hypothesis for application of the distances based on L_r -norms is that the variables are measured in the same scale. If this is not the case a standardization should be first applied.

This corresponds in particular to the use of a more general L_2 - or Euclidean distance with a metric $\mathcal{A}$ where $\mathcal{A} > 0$ (see Section 3.6)

$$d_{ij}^2 = \|x_i - x_j\|_{\mathcal{A}} = (x_i - x_j)^T \mathcal{A} (x_i - x_j). \quad (11.4)$$

L_2 -distances are given by $\mathcal{A} = \mathcal{I}_p$, but if a standardization is desired, then the weight matrix $\mathcal{A} = \text{diag}(s_{X_1 X_1}^{-1}, \dots, s_{X_p X_p}^{-1})$ may be suitable. Recall that $s_{X_k X_k}$ is the variance of the k -th component. Hence we have

$$d_{ij}^2 = \sum_{k=1}^p \frac{(x_{ik} - x_{jk})^2}{s_{X_k X_k}}. \quad (11.5)$$

Here each component has the same weight in the computation of the distances and the distances do not depend on a particular choice of the units of the measurements.

EXAMPLE 11.3 *Let us consider again the French Food expenditures. The Euclidean distance matrix (squared L_2 -norm) is*

$$D = 10^4 \cdot \begin{pmatrix} 0.00 & 5.82 & 58.19 & 3.54 & 5.15 & 151.44 & 16.91 & 36.15 & 147.99 & 51.84 & 102.56 & 271.83 \\ & 0.00 & 41.73 & 4.53 & 2.93 & 120.59 & 13.52 & 25.39 & 116.31 & 43.68 & 76.81 & 226.87 \\ & & 0.00 & 44.14 & 40.10 & 24.12 & 29.95 & 8.17 & 25.57 & 20.81 & 20.30 & 88.62 \\ & & & 0.00 & 0.76 & 127.85 & 5.62 & 21.70 & 124.98 & 31.21 & 72.97 & 231.57 \\ & & & & 0.00 & 121.05 & 5.70 & 19.85 & 118.77 & 30.82 & 67.39 & 220.72 \\ & & & & & 0.00 & 96.57 & 48.16 & 1.80 & 60.52 & 28.90 & 29.56 \\ & & & & & & 0.00 & 9.20 & 94.87 & 11.07 & 42.12 & 179.84 \\ & & & & & & & 0.00 & 46.95 & 6.17 & 18.76 & 113.03 \\ & & & & & & & & 0.00 & 61.08 & 29.62 & 31.86 \\ & & & & & & & & & 0.00 & 15.83 & 116.11 \\ & & & & & & & & & & 0.00 & 53.77 \\ & & & & & & & & & & & 0.00 \end{pmatrix}.$$

If we take the weight matrix $\mathcal{A} = \text{diag}(s_{X_1 X_1}^{-1}, \dots, s_{X_7 X_7}^{-1})$ then we obtain the distance matrix

(squared L_2 -norm)

$$D = \begin{pmatrix} 0.00 & 6.85 & 10.04 & 1.68 & 2.66 & 24.90 & 8.28 & 8.56 & 24.61 & 21.55 & 30.68 & 57.48 \\ & 0.00 & 13.11 & 6.59 & 3.75 & 20.12 & 13.13 & 12.38 & 15.88 & 31.52 & 25.65 & 46.64 \\ & & 0.00 & 8.03 & 7.27 & 4.99 & 9.27 & 3.88 & 7.46 & 14.92 & 15.08 & 26.89 \\ & & & 0.00 & 0.64 & 20.06 & 2.76 & 3.82 & 19.63 & 12.81 & 19.28 & 45.01 \\ & & & & 0.00 & 17.00 & 3.54 & 3.81 & 15.76 & 14.98 & 16.89 & 39.87 \\ & & & & & 0.00 & 17.51 & 9.79 & 1.58 & 21.32 & 11.36 & 13.40 \\ & & & & & & 0.00 & 1.80 & 17.92 & 4.39 & 9.93 & 33.61 \\ & & & & & & & 0.00 & 10.50 & 5.70 & 7.97 & 24.41 \\ & & & & & & & & 0.00 & 24.75 & 11.02 & 13.07 \\ & & & & & & & & & 0.00 & 9.13 & 29.78 \\ & & & & & & & & & & 0.00 & 9.39 \\ & & & & & & & & & & & 0.00 \end{pmatrix}. \quad (11.6)$$

When applied to contingency tables, a χ^2 -metric is suitable to compare (and cluster) rows and columns of a contingency table.

If $\mathcal{X}$ is a contingency table a row i is characterized by the conditional frequency distribution $\frac{x_{ij}}{x_{i\bullet}}$ where $x_{i\bullet} = \sum_{j=1}^p x_{ij}$ provide the marginal distributions over the rows: $\frac{x_{i\bullet}}{x_{\bullet\bullet}}$, $x_{\bullet\bullet} = \sum_{i=1}^n x_{i\bullet}$. Similarly, a column j of $\mathcal{X}$ is characterized by the conditional frequencies $\frac{x_{ij}}{x_{\bullet j}}$ where $x_{\bullet j} = \sum_{i=1}^n x_{ij}$, the marginal frequencies of the columns are $\frac{x_{\bullet j}}{x_{\bullet\bullet}}$.

A distance between two rows i_1 and i_2 is now a distance between two frequency distributions. It is common use to define this distance through the χ^2 -metric:

$$d^2(i_1, i_2) = \sum_{j=1}^p \frac{1}{\left(\frac{x_{\bullet j}}{x_{\bullet\bullet}}\right)} \left(\frac{x_{i_1 j}}{x_{i_1 \bullet}} - \frac{x_{i_2 j}}{x_{i_2 \bullet}} \right)^2. \quad (11.7)$$

Note that this can be expressed as a distance as in (11.4) between the vectors $x_1 = \left(\frac{x_{i_1 j}}{x_{i_1 \bullet}}\right)$, $x_2 = \left(\frac{x_{i_2 j}}{x_{i_2 \bullet}}\right)$ with weighting matrix $\mathcal{A} = \left\{ \text{diag} \left(\frac{x_{\bullet j}}{x_{\bullet\bullet}} \right) \right\}^{-1}$. Similarly if we are interested in clusters among the columns we have:

$$d^2(j_1, j_2) = \sum_{i=1}^n \frac{1}{\left(\frac{x_{i\bullet}}{x_{\bullet\bullet}}\right)} \left(\frac{x_{ij_1}}{x_{i\bullet j_1}} - \frac{x_{ij_2}}{x_{i\bullet j_2}} \right)^2$$

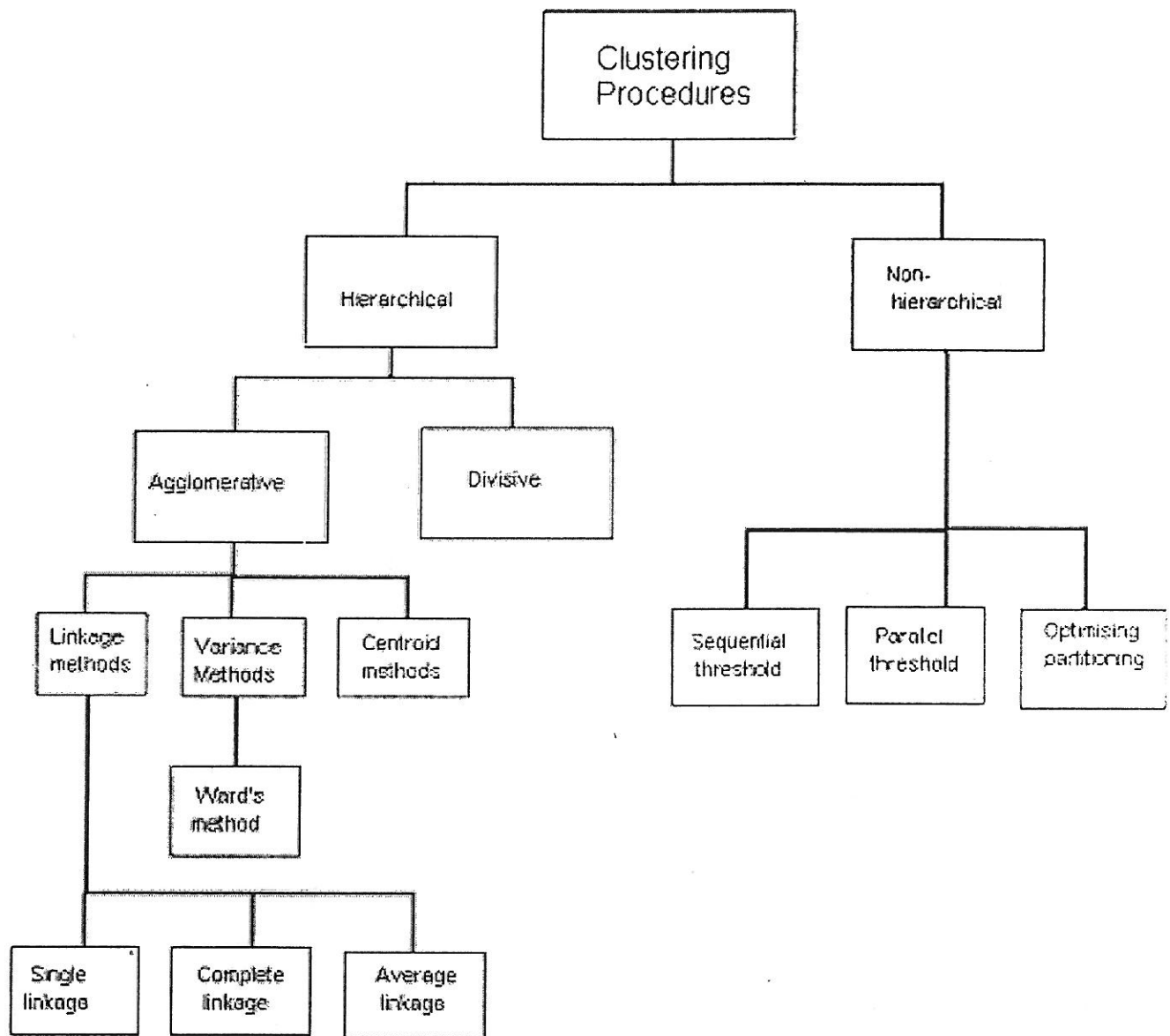
Apart from the Euclidean and the L_r -distance measures a proximity measure, the Q-correlation coefficient, can be used

$$d_{ij} = \frac{\sum_{k=1}^p (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j)}{\left\{ \sum_{k=1}^p (x_{ik} - \bar{x}_i)^2 \sum_{k=1}^p (x_{jk} - \bar{x}_j)^2 \right\}^{1/2}} \quad (11.8)$$

Note that here $\bar{x}_i$ denotes the mean over the variables $(x_{i1}, \dots, x_{ip})$.

- * The proximity between data points is measured by a distance or similarity matrix $\mathcal{D}$ whose components d_{ij} give the similarity coefficient or the distance between two points x_i, x_j .
- * There exists a variety of similarity (distance) measures for binary data (e.g., Jaccard, Tanimoto, Simple Matching coefficients) and for continuous data (e.g., L_r -distances).
- * The nature of the data could impose to choose a particular metric $\mathcal{A}$ for defining the distance (standardization, χ^2 -metric etc.).

Summary 11.2.



11.3 Cluster Algorithms

There are essentially two types of clustering methods: hierarchical algorithms and partitioning algorithms. The hierarchical algorithms can be divided further into agglomerative and splitting procedures. The first type of hierarchical clustering starts from the finest partition possible (each observation forms a cluster) and puts groups together into clusters. The second type starts with the coarsest partition possible: the one cluster that contains all observations. It then proceeds by splitting up this group into clusters of smaller size.

The partitioning algorithms start from a given group definition and proceed by exchanging elements between groups until a certain score is optimized. The main difference between the two clustering techniques is that once groups are found and elements are assigned to these groups in hierarchical clustering this assignment cannot be changed. In partitioning techniques the assignment of objects into groups may well change during the algorithm.

Hierarchical Algorithms, Agglomerative Techniques

Agglomerative algorithms are used very frequently in practice. The algorithm consists of the following steps:

1. Construct the finest partition.
2. Compute the distance matrix $\mathcal{D}$.

DO

3. Find the clusters with the closest distance.
4. Put those two clusters into one cluster.
5. Compute the distance between the new groups and obtain a reduced distance matrix $\mathcal{D}$.

UNTIL all clusters are agglomerated into $\mathcal{X}$.

If two objects or groups P and Q are to be united one obtains the distance to another group (object) R by the following distance function

$$d(R, P + Q) = \delta_1 d(R, P) + \delta_2 d(R, Q) + \delta_3 d(P, Q) + \delta_4 |d(R, P) - d(R, Q)|. \quad (11.9)$$

The δ_j 's are weighting factors that lead to different agglomerative algorithms as described in Table 11.2. Here $n_P = \sum_{i=1}^n \mathbf{I}(x_i \in P)$ is the number of objects in group P . The values of n_Q and n_R are defined in analogy.

Name	δ_1	δ_2	δ_3	δ_4
Single linkage	1/2	1/2	0	-1/2
Complete linkage	1/2	1/2	0	1/2
Average linkage (unweighted)	1/2	1/2	0	0
Average linkage (weighted)	$\frac{n_P}{n_P + n_Q}$	$\frac{n_Q}{n_P + n_Q}$	0	0
Centroid	$\frac{n_P}{n_P + n_Q}$	$\frac{n_Q}{n_P + n_Q}$	$-\frac{n_P n_Q}{(n_P + n_Q)^2}$	0
Median	1/2	1/2	-1/4	0
Ward	$\frac{n_R + n_P}{n_R + n_P + n_Q}$	$\frac{n_R + n_Q}{n_R + n_P + n_Q}$	$-\frac{n_R}{n_R + n_P + n_Q}$	0

Table 11.2. Computations of group distances.

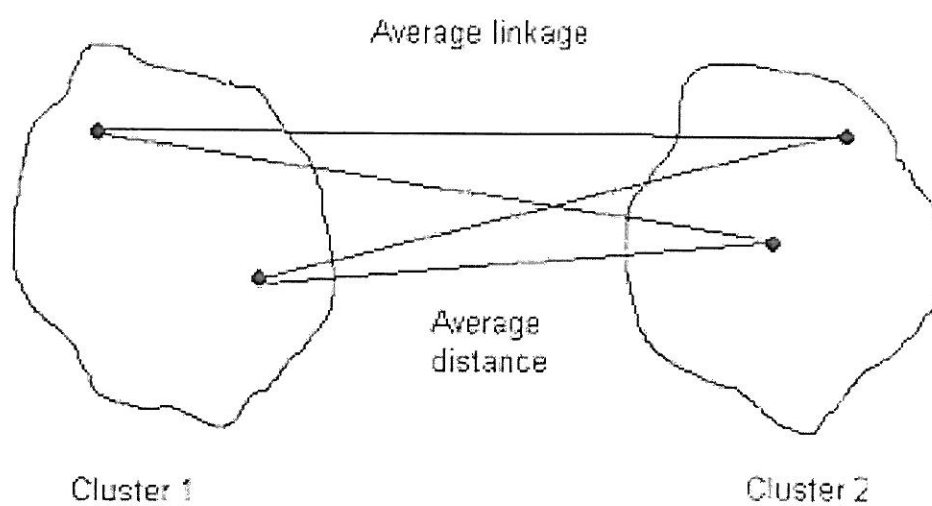
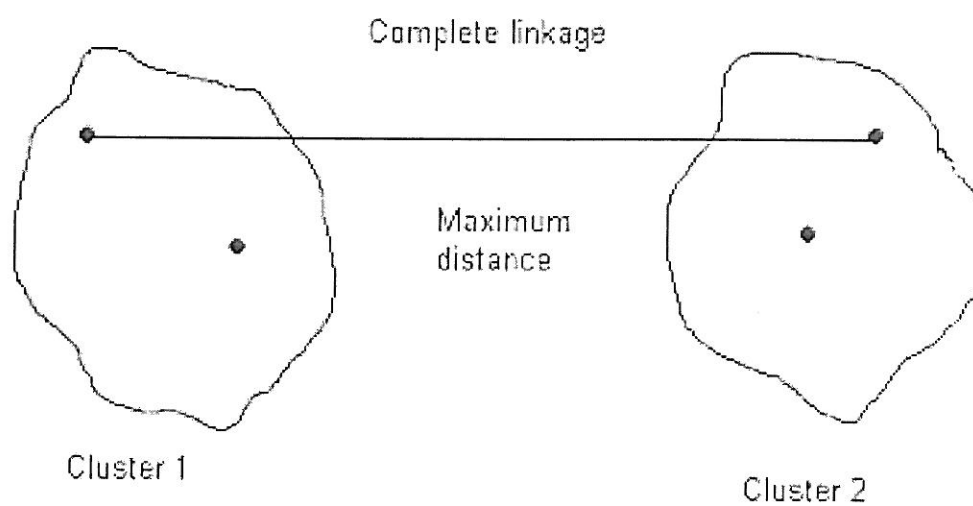
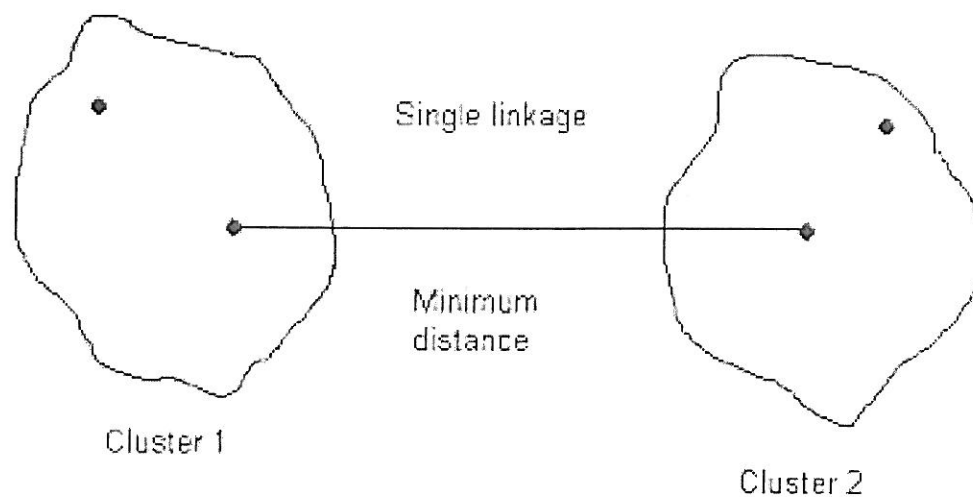
EXAMPLE 11.4 Let us examine this for the three point example $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (5, 5)$ and the squared Euclidean distance matrix with single linkage weighting. The algorithm starts with $N = 3$ clusters $P = \{x_1\}$, $Q = \{x_2\}$, $R = \{x_3\}$. The distance matrix D_2 is defined in Example 11.2. The smallest distance in D_2 is that between the clusters P and Q so by definition of step 4 in the above algorithm we put together these clusters into $P + Q = \{x_1, x_2\}$. The single linkage distance between the remaining two clusters is by Table 11.2 and (11.2) equal to

$$\begin{aligned}
 d(R, P + Q) &= \frac{1}{2}d(R, P) + \frac{1}{2}d(R, Q) - \frac{1}{2}|d(R, P) - d(R, Q)| \quad (11.10) \\
 &= \frac{1}{2}d_{13} + \frac{1}{2}d_{23} - \frac{1}{2} \cdot |d_{13} - d_{23}| \\
 &= \frac{50}{2} + \frac{41}{2} - \frac{1}{2} \cdot |50 - 41| \\
 &= 41
 \end{aligned}$$

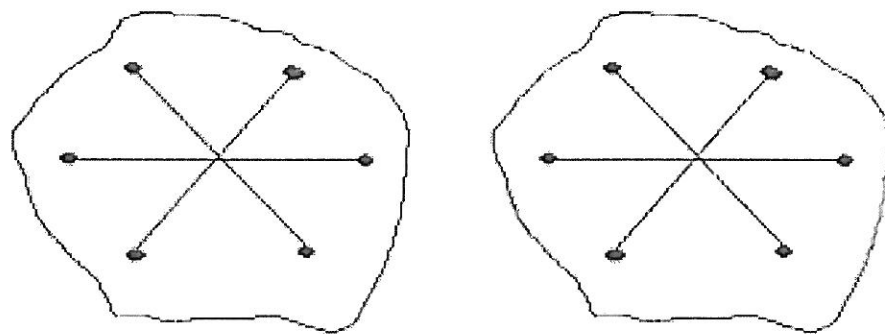
The reduced distance matrix is then $\begin{pmatrix} 0 & 41 \\ 41 & 0 \end{pmatrix}$. The next and last step is to unite the clusters R and $P + Q$ into a single cluster $\mathcal{X}$, the original data matrix.

For more data points than in the example above a visualization of the implication of clusters is desirable. A graphical representation of the sequence of clustering is the so-called dendrogram. It displays the observations, the sequence of clusters and the distances between the clusters. On the vertical axis the index of points is given; the horizontal axis gives the distance between the clusters.

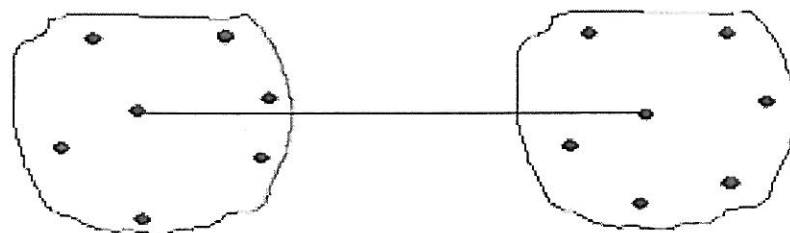
High level of the latter indicates the clustering of heterogeneous groups. So if we choose to "cut the tree" at a desired level, the cut branches on the left describe the corresponding clusters.

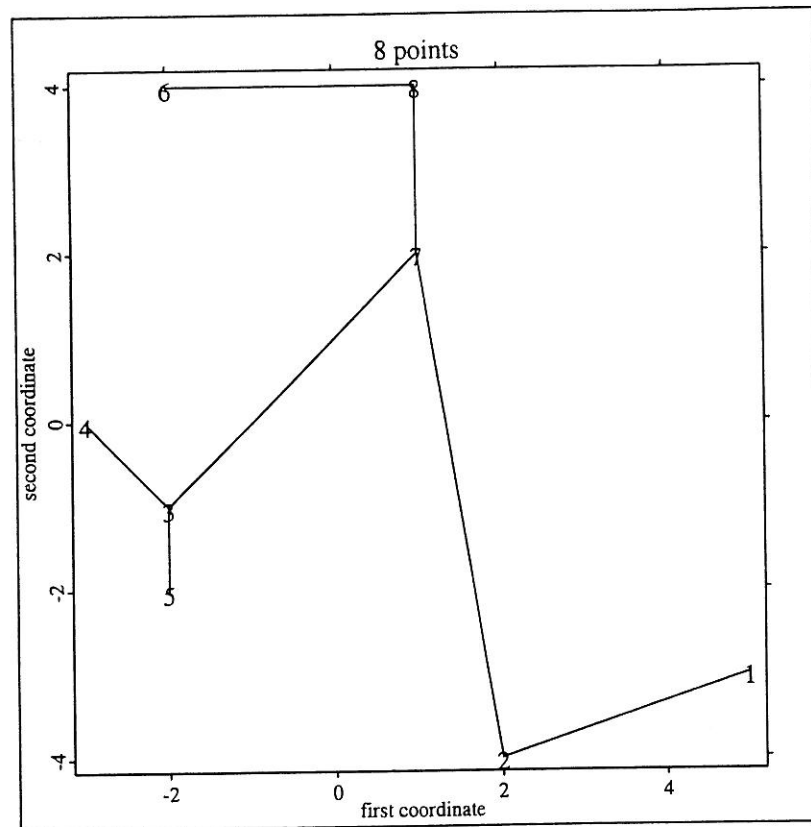


Ward's method



Centroid method



Figure 11.1. The 8 points example. $\rightarrow$ c.xp1lus8p

EXAMPLE 11.5 In the following we describe the single linkage algorithm for the eight data points displayed in Figure 11.1.

Then the distance matrix (L_2 distances) is

$$D = \begin{pmatrix} 0 & 10 & 53 & 73 & 50 & 98 & 41 & 65 \\ & 0 & 25 & 41 & 20 & 80 & 37 & 65 \\ & & 0 & 2 & 1 & 25 & 18 & 34 \\ & & & 0 & 5 & 17 & 20 & 32 \\ & & & & 0 & 36 & 25 & 45 \\ & & & & & 0 & 13 & 9 \\ & & & & & & 0 & 4 \\ & & & & & & & 0 \end{pmatrix}$$

and the dendrogram is shown in Figure 11.2.

If we decide to cut the tree at the level 10 we define three clusters: $\{1, 2\}$, $\{3, 4, 5\}$ and $\{6, 7, 8\}$.

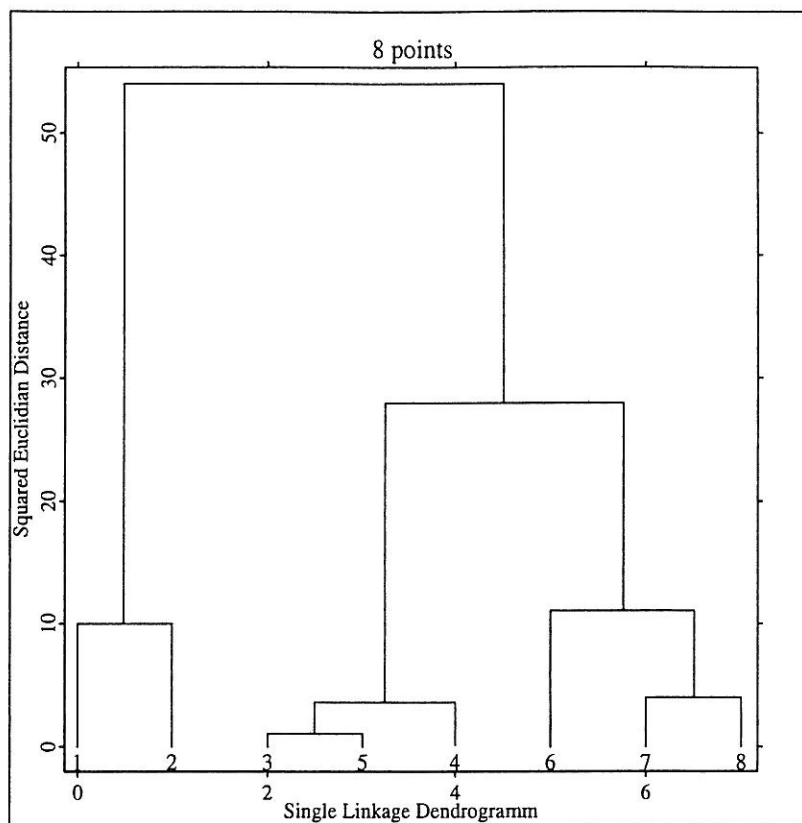


Figure 11.2. The dendrogram for the 8 points example, Single linkage algorithm. $\rightarrow c.xp11us8p$

The *single linkage* algorithm defines as the distance between two groups the smallest value of the individual distances. From Table 11.2 we see indeed that in this case

$$d(R, P + Q) = \min\{d(R, P), d(R, Q)\}. \quad (11.11)$$

It is also called the *Nearest Neighbor* algorithm. A consequence of this construction is that single linkage tends to build large groups. Not so well separated groups although different may thus be classified into one group as far as they have two approximate points. The *complete linkage* algorithm tries to correct this by considering the largest (individual) distances, indeed from Table 11.2 we see that the complete linkage distance can be written as

$$d(R, P + Q) = \max\{d(R, P), d(R, Q)\}. \quad (11.12)$$

It is also called *Farthest Neighbor* algorithm. This algorithm will cluster groups where all the points are proximate, since it compares the largest distances. The *average linkage* algorithm (weighted or unweighted) proposes a compromise between the two preceding, since

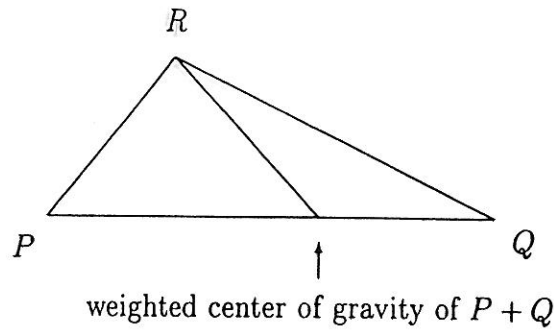


Figure 11.3.

it computes an average distance. The *centroid* algorithm is very similar and uses the natural geometrical distance between R and the weighted center of gravity of P and Q (see Figure 11.3).

The *Ward clustering* algorithm computes the distance between groups according to the formula in Table 11.2. The main difference between this algorithm and the linkage procedures consists in the unification procedure. The Ward algorithm does not put together groups with smallest distance, rather it joins groups that do not increase too much a given measure of heterogeneity. The aim of the Ward procedure is to unify groups such that the variation inside these groups is not increased too drastically: the resulting groups are as homogeneous as possible.

As a measure of heterogeneity for a group R the inertia inside the group is used. This inertia is defined as follows:

$$I_R = \sum_{i=1}^{n_R} d^2(x_i, \bar{x}_R) \quad (11.13)$$

where $\bar{x}_R$ is the center of gravity (mean) of the groups. I_R clearly provides a scalar measure of the dispersion of the group around its center of gravity. If the usual Euclidean distance is used, it represents the sum of the variances of the p components of x_i inside the group R .

When two objects or groups P and Q will be joined, the new group $P + Q$ will have a larger inertia I_{P+Q} . It can be shown that the corresponding increase of inertia is given by

$$\Delta(P, Q) = \frac{n_P n_Q}{n_P + n_Q} d^2(P, Q). \quad (11.14)$$

The Ward algorithm can be redefined as "Join these groups that give the smallest increase of $\Delta(P, Q)$ ". It is easy to prove that when P and Q are joined, the new criterion values are indeed given by (11.9) along with the values of δ_i given in Table 11.2, when the centroid formula is used to modify $d^2(R, P + Q)$. So, the Ward algorithm is related to the centroid algorithm, but rather with an "inertial" distance Δ than the "geometric" distance d^2 .

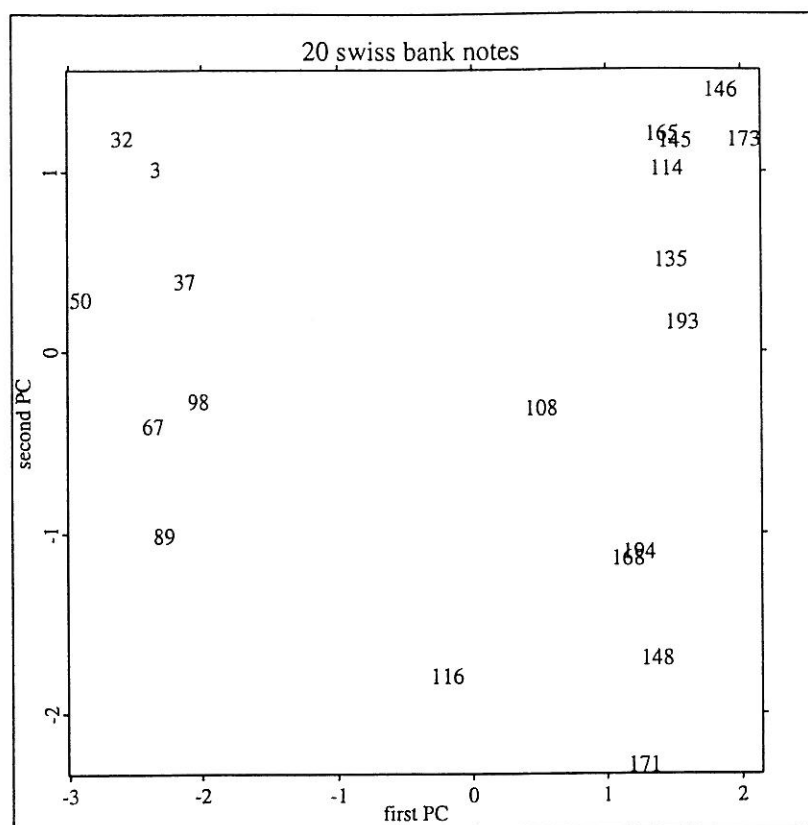


Figure 11.4. PCA for 20 randomly chosen bank notes. $\rightarrow$ c.xp11usbank

As pointed out in Section 11.2 all the algorithms above can be particularized by the choice of the metric $\mathcal{A}$ defining the geometric distance d^2 . If the results of a clustering algorithm are illustrated on graphical representations of individuals in spaces of small dimension (by principal components - normalized or not - or by correspondence analysis for contingency tables) it is important to be coherent in the choice of the metric used.

EXAMPLE 11.6 *As an example we take 20 random observations out of the bank notes and apply the Ward technique for Euclidean distances. Figure 11.4 shows the first two PCs of these data, Figure 11.5 displays the dendrogram.*

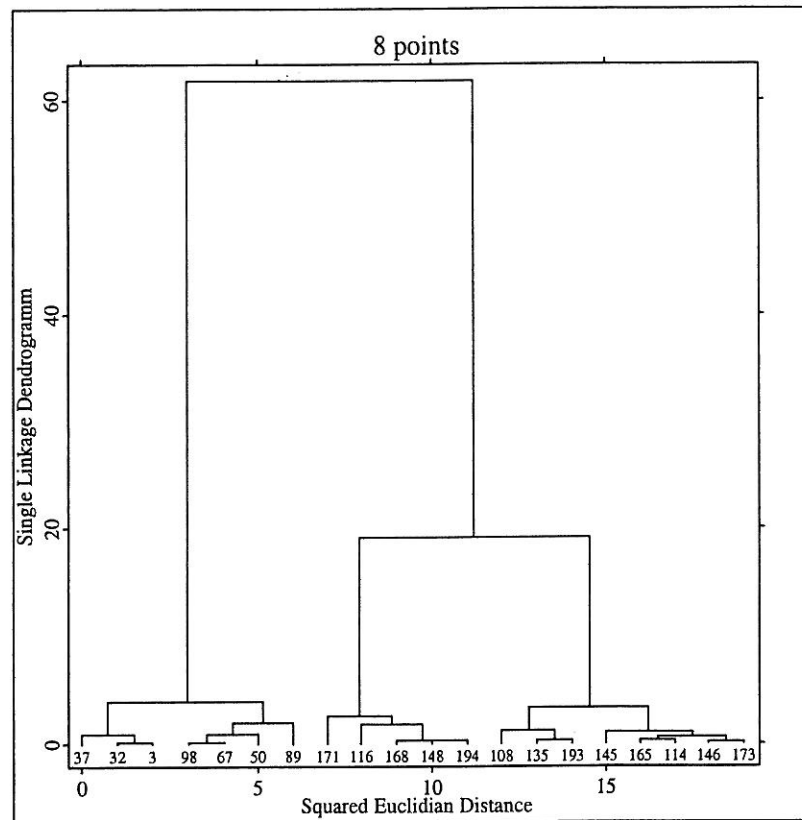


Figure 11.5. The dendrogram for the 20 bank notes, Ward algorithm.
→c.xpplusbank

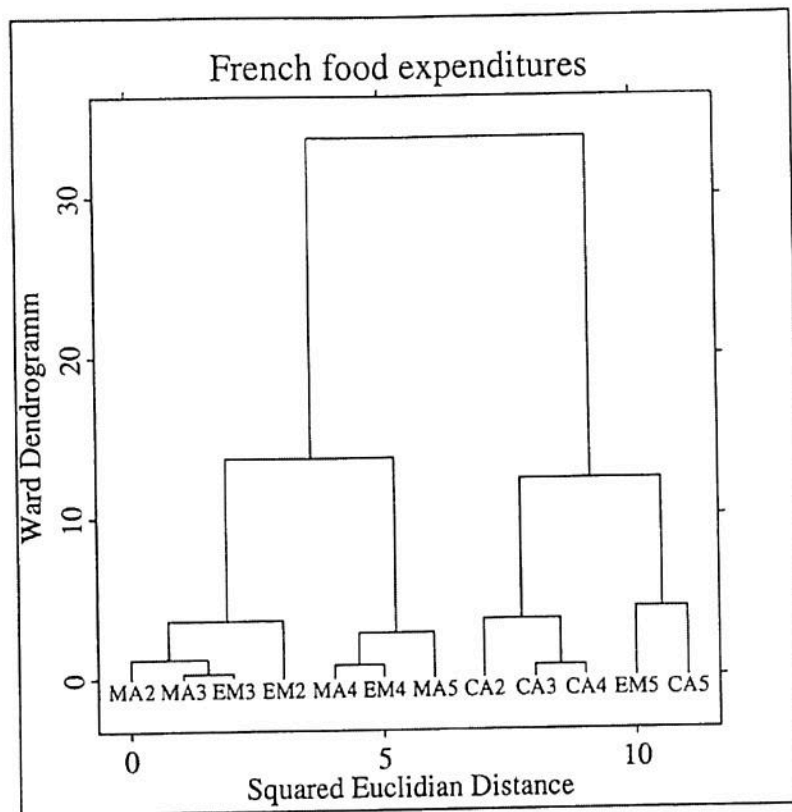


Figure 11.6. The dendrogram for the French food expenditures, Ward algorithm. $\rightarrow$ c.xp11usfood

EXAMPLE 11.7 Let us consider now the French Food expenditures. Here we use standardized data as in Chapter 9 which is equivalent to use $A = \text{diag}(s_{X_1X_1}^{-1}, \dots, s_{X_7X_7}^{-1})$ as weight matrix in the L_2 -distance. The NPCA plot of the individuals was given in Figure 9.7. The Euclidean distance matrix is of course that of (11.6). The dendrogram obtained by using the Ward algorithm is shown in Figure 11.6.

If we would aim to have only two groups we see from Figure 11.6 that these are $\{CA2, CA3, CA4, CA5, EM5\}$ and $\{MA2, MA3, MA4, MA5, EM2, EM3, EM4\}$. Clustering three groups is somewhat arbitrary (the levels of distances are very similar). If we would choose four groups we obtain $\{CA2, CA3, CA4\}$, $\{EM2, MA2, EM3, MA3\}$, $\{EM4, MA4, MA5\}$ and $\{EM5, CA5\}$. This shows a balance between socio-professional levels and size of the families to determine the clusters. The four groups are clearly well represented in the NPCA plot in Figure 9.7.

- * The class of clustering algorithms can be divided into two types: hierarchical and partitioning algorithms. Hierarchical algorithms start with the finest (coarsest) possible partition and put groups together (splits groups) from step to step. Partitioning algorithms start from a preliminary clustering and exchange group elements until a certain score is reached.
- * Hierarchical agglomerative techniques are frequently used in practice. They start from the finest possible structure (all data points form clusters), compute the distance matrix for these clusters and join the clusters with the smallest distance. This step is repeated until all points are united in one cluster.
- * The agglomerative procedure depends on the definition of the distance between two clusters. Often used distances are single linkage, complete linkage, Ward distance.
- * The process of the unification of clusters can be graphically represented by a dendrogram.

Summary 11.3.

A.5 French Food Data

The data set consists of average expenditures for food of several types of families in France (manual workers = MA, employees = EM, managers = CA) with different numbers of children (2,3,4 or 5 children). The data stem from Lebart, Morineau and F  nelon (1982).

		bread	vegetables	fruits	meat	poultry	milk	wine
1	MA2	332	428	354	1437	526	247	427
2	EM2	293	559	388	1527	567	239	258
3	CA2	372	767	562	1948	927	235	433
4	MA3	406	563	341	1507	544	324	407
5	EM3	386	608	396	1501	558	319	363
6	CA3	438	843	689	2345	1148	243	341
7	MA4	534	660	367	1620	638	414	407
8	EM4	460	699	484	1856	762	400	416
9	CA4	385	789	621	2366	1149	304	282
10	MA5	655	776	423	1848	759	495	486
11	EM5	584	995	548	2056	893	518	319
12	CA5	515	1097	887	2630	1167	561	284
	$\bar{x}$	446,7	737,8	505,0	1886,7	803,2	358,2	368,6
	$s_{X_i X_i}$	102,6	172,2	158,1	378,9	238,9	112,1	68,7

Table A.5. French food expenditures.

A.6 Car Marks

The data are averaged marks for 24 car types from a sample of 40 persons. The marks range from 1 (very good) to 6 (very bad) like German school marks. The variables are:

- X_1 : A Economy,
- X_2 : B Service,
- X_3 : C Non-depreciation of value,
- X_4 : D Price, Mark 1 for very cheap cars
- X_5 : E Design,
- X_6 : F Sporty car,
- X_7 : G Safety,
- X_8 : H Easy handling.

	Type	Model	Economy	Service	Value	Price	Design	Sport.	Safety	Easy h.
1	Audi	100	3.9	2.8	2.2	4.2	3.0	3.1	2.4	2.8
2	BMW	5 series	4.8	1.6	1.9	5.0	2.0	2.5	1.6	2.8
3	Citroen	AX	3.0	3.8	3.8	2.7	4.0	4.4	4.0	2.6
4	Ferrari		5.3	2.9	2.2	5.9	1.7	1.1	3.3	4.3
5	Fiat	Uno	2.1	3.9	4.0	2.6	4.5	4.4	4.4	2.2
6	Ford	Fiesta	2.3	3.1	3.4	2.6	3.2	3.3	3.6	2.8
7	Hyundai		2.5	3.4	3.2	2.2	3.3	3.3	3.3	2.4
8	Jaguar		4.6	2.4	1.6	5.5	1.3	1.6	2.8	3.6
9	Lada	Samara	3.2	3.9	4.3	2.0	4.3	4.5	4.7	2.9
10	Mazda	323	2.6	3.3	3.7	2.8	3.7	3.0	3.7	3.1
11	Mercedes	200	4.1	1.7	1.8	4.6	2.4	3.2	1.4	2.4
12	Mitsubishi	Galant	3.2	2.9	3.2	3.5	3.1	3.1	2.9	2.6
13	Nissan	Sunny	2.6	3.3	3.9	2.1	3.5	3.9	3.8	2.4
14	Opel	Corsa	2.2	2.4	3.0	2.6	3.2	4.0	2.9	2.4
15	Opel	Vectra	3.1	2.6	2.3	3.6	2.8	2.9	2.4	2.4
16	Peugeot	306	2.9	3.5	3.6	2.8	3.2	3.8	3.2	2.6
17	Renault	19	2.7	3.3	3.4	3.0	3.1	3.4	3.0	2.7
18	Rover		3.9	2.8	2.6	4.0	2.6	3.0	3.2	3.0
19	Toyota	Corolla	2.5	2.9	3.4	3.0	3.2	3.1	3.2	2.8
20	Volvo		3.8	2.3	1.9	4.2	3.1	3.6	1.6	2.4
21	Trabant	601	3.6	4.7	5.5	1.5	4.1	5.8	5.9	3.1
22	VW	Golf	2.4	2.1	2.0	2.6	3.2	3.1	3.1	1.6
23	VW	Passat	3.1	2.2	2.1	3.2	3.5	3.5	2.8	1.8
24	Wartburg	1.3	3.7	4.7	5.5	1.7	4.8	5.2	5.5	4.0

Table A.6. Averaged car marks.

A.7 French Baccalauréat Frequencies

The data come from observations of 202100 baccalauréats from France in 1976 and give the frequencies for different sets of modalities classified into regions. For a reference see Bourouche and Saporta (1980). The variables (modalities) are:

- X_1 : A Philosophy-Letters,
- X_2 : B Economics and Social Sciences,
- X_3 : C Mathematics and Physics,
- X_4 : D Mathematics and Natural Sciences,
- X_5 : E Mathematics and Techniques,
- X_6 : F Industrial Techniques,
- X_7 : G Economic Techniques,
- X_8 : H Computer Techniques.

Abbrev.	Region	A	B	C	D	E	F	G	H	total
1	ILDF	9724	5650	8679	9432	839	3353	5355	83	43115
2	CHAM	924	464	567	984	132	423	736	12	4242
3	PICA	1081	490	830	1222	118	410	743	13	4907
4	HNOR	1135	587	686	904	83	629	813	13	4850
5	CENT	1482	667	1020	1535	173	629	989	26	6521
6	BNOR	1033	509	553	1063	100	433	742	13	4446
7	BOUR	1272	527	861	1116	219	769	1232	13	6009
8	NOPC	2549	1141	2164	2752	587	1660	1951	41	12845
9	LORR	1828	681	1364	1741	302	1289	1683	15	8903
10	ALSA	1076	443	880	1121	145	917	1091	15	5688
11	FRAC	827	333	481	892	137	451	618	18	3757
12	PAYL	2213	809	1439	2623	269	990	1783	14	10140
13	BRET	2158	1271	1633	2352	350	950	1509	22	10245
14	PCHA	1358	503	639	1377	164	495	959	10	5505
15	AQUI	2757	873	1466	2296	215	789	1459	17	9872
16	MIDI	2493	1120	1494	2329	254	855	1565	28	10138
17	LIMO	551	297	386	663	67	334	378	12	2688
18	RHOA	3951	2127	3218	4743	545	2072	3018	36	19710
19	AUVE	1066	579	724	1239	126	476	649	12	4871
20	LARO	1844	816	1154	1839	156	469	993	16	7287
21	PROV	3944	1645	2415	3616	343	1236	2404	22	15625
22	CORS	327	31	85	178	9	27	79	0	736
total		45593	21563	32738	46017	5333	19656	30749	451	202100

Table A.7. French baccalauréats.